

## HKL ICA Matrix in the Bolt Pipeline

Fact-checked against the current trainer and affine-node code.

**Short answer.** Yes, conceptually:

PCA  $\rightarrow$  ICA  $\rightarrow$  component mask  $\rightarrow$  inverse ICA  $\rightarrow$  PCA back-projection.

The “identity matrix to remove components” is a diagonal mask  $D$ : artifact ICs get 0, kept ICs get 1.

**Column-vector form, full-rank ICA.** For one channel vector  $x \in \mathbb{R}^n$ :

$$x_{\text{out}} = Cx, \quad C = P^\top ADWP.$$

|          |                                                           |
|----------|-----------------------------------------------------------|
| $P$      | retained PCA projection                                   |
| $W$      | ICA unmixing matrix                                       |
| $D$      | diagonal keep/remove mask for ICA components              |
| $A$      | ICA mixing matrix, i.e. inverse ICA in the retained space |
| $P^\top$ | PCA back-projection                                       |

**Bolt row-vector form.** Bolt applies row samples by right multiplication:

$$X_{\text{out}} = XM, \quad M = C^\top.$$

The trained-channel block is embedded into a full-stream affine matrix; untrained channels are identity.